

Eastern Washington University
College of Science, Technology, Engineering, and Mathematics
Department of Mathematics

MATH 241: Multivariable Calculus
Dr. Nievergelt
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Integrals with Toroidal Coordinates #6

Rafael Faria
(Student ID: 00989433)

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Problem 1

With $R > 0$ fixed, define *toroidal* coordinates by

$$X(\theta, \lambda, \sigma) = [R + \sigma \cdot \cos(\lambda)] \cdot \cos(\theta),$$

$$Y(\theta, \lambda, \sigma) = [R + \sigma \cdot \cos(\lambda)] \cdot \sin(\theta),$$

$$Z(\theta, \lambda, \sigma) = \sigma \cdot \sin(\lambda).$$

The longitude θ and latitude λ parametrize the surface of a torus, and with σ , a solid doughnut $\mathcal{D}$:

$$-\pi \leq \theta \leq \pi, \quad -\pi \leq \lambda \leq \pi, \quad 0 \leq \sigma \leq S < R.$$

Question: Calculate the average of $f(x, y) = x^2 + y^2$ over the solid doughnut $\mathcal{D}$. **Solution:** To calculate the average of f over $\mathcal{D}$, we will calculate the following:

$$f_{ave} = \frac{\int \int \int_{\mathcal{D}} f(x, y) \cdot dV}{\int \int \int_{\mathcal{D}} 1 \cdot dV}$$

But to do so we will first calculate dV in the following manner:

$$dV = \det \left(\left| \frac{\partial(x, y, z)}{\partial(\theta, \lambda, \sigma)} \right| \right)$$

$$dV = \det \left(\begin{bmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \lambda} & \frac{\partial x}{\partial \sigma} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \lambda} & \frac{\partial y}{\partial \sigma} \\ \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \lambda} & \frac{\partial z}{\partial \sigma} \end{bmatrix} \right)$$

$$\frac{\partial x}{\partial \theta} = -[R + \sigma \cdot \cos(\lambda)] \cdot \sin(\theta), \quad \frac{\partial x}{\partial \lambda} = -\sigma \sin(\lambda) \cos(\theta), \quad \frac{\partial x}{\partial \sigma} = \cos(\lambda) \cos(\theta)$$

$$\frac{\partial y}{\partial \theta} = [R + \sigma \cdot \cos(\lambda)] \cos(\theta), \quad \frac{\partial y}{\partial \lambda} = -\sigma \sin(\lambda) \sin(\theta), \quad \frac{\partial y}{\partial \sigma} = \cos(\lambda) \sin(\theta)$$

$$\frac{\partial z}{\partial \theta} = 0, \quad \frac{\partial z}{\partial \lambda} = \sigma \cos(\lambda), \quad \frac{\partial z}{\partial \sigma} = \sin(\lambda)$$

$$dV = \det \left(\begin{bmatrix} -[R + \sigma \cdot \cos(\lambda)] \cdot \sin(\theta) & -\sigma \sin(\lambda) \cos(\theta) & \cos(\lambda) \cos(\theta) \\ [R + \sigma \cdot \cos(\lambda)] \cos(\theta) & -\sigma \sin(\lambda) \sin(\theta) & \cos(\lambda) \sin(\theta) \\ 0 & \sigma \cos(\lambda) & \sin(\lambda) \end{bmatrix} \right)$$

Now we can utilize the zero entry in the matrix by using cofactor expansion to calculate the determinant of the matrix in the following way:

$$dV = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$$

$$dV = a_{31}((-1)^{3+1} \cdot M_{31}) + a_{32}((-1)^{3+2} \cdot M_{32}) + a_{33}((-1)^{3+3} \cdot M_{33})$$

$$dV = a_{31}M_{31} - a_{32}M_{32} + a_{33}M_{33}$$

$$dV = 0M_{31} - \sigma \cos(\lambda)M_{32} + \sin(\lambda)M_{33}$$

$$dV = \sin(\lambda)M_{33} - \sigma \cos(\lambda)M_{32}$$

Now we compute M_{32} and M_{33} which are the determinants of the 2×2 matrices that result from eliminating the 3rd row and second and third columns respectively:

$$\begin{aligned} M_{32} &= \det \begin{pmatrix} -[R + \sigma \cdot \cos(\lambda)] \cdot \sin(\theta) & \cos(\lambda) \cos(\theta) \\ [R + \sigma \cdot \cos(\lambda)] \cos(\theta) & \cos(\lambda) \sin(\theta) \end{pmatrix} \\ &= -\cos(\lambda) \sin^2(\theta)[R + \sigma \cos(\lambda)] - \cos(\lambda) \cos^2(\theta)[R + \sigma \cos(\lambda)] \\ &= -\cos(\lambda)[R + \sigma \cos(\lambda)](\sin^2(\theta) + \cos^2(\theta)) \\ M_{32} &= -[R + \sigma \cos(\lambda)] \cos(\lambda) \end{aligned}$$

$$\begin{aligned} M_{33} &= \det \begin{pmatrix} -[R + \sigma \cdot \cos(\lambda)] \cdot \sin(\theta) & -\sigma \sin(\lambda) \cos(\theta) \\ [R + \sigma \cdot \cos(\lambda)] \cos(\theta) & -\sigma \sin(\lambda) \sin(\theta) \end{pmatrix} \\ &= \sigma \sin(\lambda) \sin^2(\theta)[R + \sigma \cos(\lambda)] + \sigma \sin(\lambda) \cos^2(\theta)[R + \sigma \cos(\lambda)] \\ &= \sigma[R + \sigma \cos(\lambda)](\sin^2(\theta) + \cos^2(\theta)) \\ M_{33} &= \sigma[R + \sigma \cos(\lambda)] \end{aligned}$$

Now we can compute dV as follows:

$$\begin{aligned} dV &= \sin(\lambda)(\sigma[R + \sigma \cos(\lambda)]) - \sigma \cos(\lambda)(-\cos(\lambda)[R + \sigma \cos(\lambda)]) \\ &= \sin(\lambda)(R\sigma \sin(\lambda) + \sigma^2 \sin(\lambda) \cos(\lambda)) - \sigma \cos(\lambda)(-R \cos(\lambda) - \sigma \cos^2(\lambda)) \\ &= R\sigma \sin^2(\lambda) + \sigma^2 \sin^2(\lambda) \cos(\lambda) + R\sigma \cos^2(\lambda) + \sigma^2 \cos^3(\lambda) \\ &= R\sigma(\sin^2(\lambda) + \cos^2(\lambda)) + \sigma^2 \cos(\lambda)(\sin^2(\lambda) + \cos^2(\lambda)) \\ dV &= R\sigma + \sigma^2 \cos(\lambda) \end{aligned}$$

$$dV = [R + \sigma \cos(\lambda)]\sigma$$

While our calculated dV accounts for change of coordinates, it is missing the differentials of the new coordinate system. Therefore,

$$dV = [R + \sigma \cos(\lambda)]\sigma \cdot d\sigma d\lambda d\theta$$

Now we can calculate the average value of $f(x, y)$ over the Doughnut $\mathcal{D}$ as parametrized by σ, λ , and θ in the following manner:

$$\begin{aligned} f(X, Y) &= ([R + \sigma \cos(\lambda)] \cos(\theta))^2 + ([R + \sigma \cos(\lambda)] \sin(\theta))^2 \\ f(X, Y) &= R^2 + 2R\sigma \cos(\lambda) + \sigma^2 \cos^2(\lambda) \\ f(X, Y) &= [R + \sigma \cos(\lambda)]^2 \\ f_{ave} &= \frac{\int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S [R + \sigma \cos(\lambda)]^2 \cdot [R + \sigma \cos(\lambda)]\sigma \cdot d\sigma d\lambda d\theta}{\int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S [R + \sigma \cos(\lambda)]\sigma \cdot d\sigma d\lambda d\theta} = \frac{I}{V} \end{aligned}$$

For now we will focus on the integral I in the numerator of f_{ave} :

$$\begin{aligned} I &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S [R + \sigma \cos(\lambda)]^2 \cdot [R + \sigma \cos(\lambda)]\sigma \cdot d\sigma d\lambda d\theta \\ &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S \sigma [R + \sigma \cos(\lambda)]^3 \cdot d\sigma d\lambda d\theta \\ &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S R^3 \sigma + 3R^2 \sigma^2 \cos(\lambda) + 3R\sigma^3 \cos^2(\lambda) + \sigma^4 \cos^3(\lambda) \cdot d\sigma d\lambda d\theta \\ &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \left[\frac{R^3}{2} \sigma^2 + R^2 \sigma^3 \cos(\lambda) + \frac{3R}{4} \sigma^4 \cos^2(\lambda) + \frac{\sigma^5}{5} \cos^3(\lambda) \right]_0^S d\lambda d\theta \\ &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{S^2 R^3}{2} + S^3 R^2 \cos(\lambda) + \frac{3RS^4}{4} \cos^2(\lambda) + \frac{S^5}{5} \cos^3(\lambda) \cdot d\lambda d\theta \\ &= \int_{-\pi}^{\pi} \left[\frac{S^2 R^3}{2} \lambda + S^3 R^2 \sin(\lambda) + \frac{3S^4 R}{8} \cos(\lambda) \sin(\lambda) + \frac{3S^4 R}{8} \lambda + \frac{S^5}{15} \cos^2(\lambda) \sin(\lambda) + \frac{2S^5}{15} \sin(\lambda) \right]_{-\pi}^{\pi} d\theta \\ &= \int_{-\pi}^{\pi} \frac{\pi}{2} S^2 R^3 + \frac{3\pi}{8} RS^4 - \left(-\frac{\pi}{2} S^2 R^3 - \frac{3\pi}{8} S^4 R \right) d\theta \\ &= \int_{-\pi}^{\pi} S^2 R^3 \pi + \frac{3}{4} RS^4 \pi d\theta \\ &= \left[S^2 R^3 \pi \theta + \frac{3}{4} RS^4 \pi \theta \right]_{-\pi}^{\pi} \\ I &= S^2 R^3 \pi^2 + \frac{3}{4} RS^4 \pi^2 - (-S^2 R^3 \pi^2 - \frac{3}{4} RS^4 \pi^2) \end{aligned}$$

$$I = 2S^2R^3\pi^2 + \frac{3}{2}S^4R\pi^2$$

Now we will calculate the integral V from the denominator of f_{ave} :

$$\begin{aligned}
 V &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S [R + \sigma \cos(\lambda)] \sigma \cdot d\sigma d\lambda d\theta \\
 &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_0^S R\sigma + \sigma^2 \cos(\lambda) \quad d\sigma d\lambda d\theta \\
 &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \left[\frac{R}{2} \sigma^2 + \frac{\sigma^3}{3} \cos(\lambda) \right]_0^S \quad d\lambda d\theta \\
 &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{RS^2}{2} + \frac{S^3}{3} \cos(\lambda) \quad d\lambda d\theta \\
 &= \int_{-\pi}^{\pi} \left[\frac{RS^2}{2} \lambda + \frac{S^3}{3} \sin(\lambda) \right]_{-\pi}^{\pi} \quad d\theta \\
 &= \int_{-\pi}^{\pi} \frac{\pi}{2} RS^2 + \frac{\pi}{2} RS^2 \quad d\theta \\
 &= \int_{-\pi}^{\pi} \pi RS^2 \quad d\theta \\
 &= [\pi RS^2 \theta]_{-\pi}^{\pi} \\
 V &= 2\pi^2 RS^2
 \end{aligned}$$

Now we can calculate the average of $f(x, y)$ over $\mathcal{D}$:

$$\begin{aligned}
 f_{ave} &= \frac{I}{V} = \frac{2S^2R^3\pi^2 + \frac{3}{2}RS^4\pi^2}{2\pi^2RS^2} = \frac{S^2R^3\pi^2}{\pi^2RS^2} + \frac{3}{4} \frac{RS^4\pi^2}{\pi^2RS^2} \\
 f_{ave} &= R^2 + \frac{3}{4}S^2
 \end{aligned}$$

Therefore, the average value of $f(x, y)$ over the Doughnut $\mathcal{D}$ parametrized by the values of σ, λ , and θ is:

$$\boxed{f_{ave} = R^2 + \frac{3}{4}S^2}$$